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(54) **ELECTRONIC APPARATUS FOR STABLE ELECTRICAL CONNECTION**

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See application file for complete search history.

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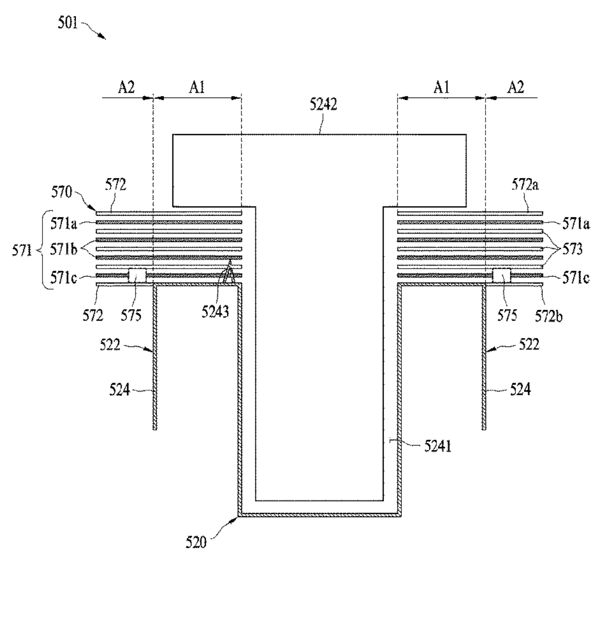
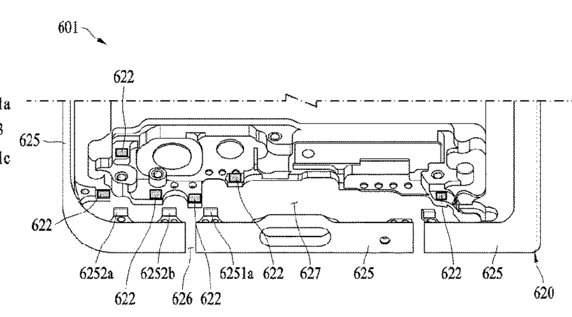
(57) **ABSTRACT**

An electronic apparatus includes: a housing including a coupling region; and a printed circuit board (PCB) including: an overlap region that overlaps the coupling region; a non-overlap region that does not overlap the coupling region; a plurality of metal layers including a plurality of lines; and a void portion in which a portion of at least one metal layer, among the plurality of metal layers, is not formed so that the at least one metal layer is discontinuous or terminated in the void portion, the at least one metal layer including a metal layer that is closest to the coupling region among the plurality of metal layers.

15 Claims, 12 Drawing Sheets





(51) **Int. Cl.**

H04M 1/02	(2006.01)
H05K 1/02	(2006.01)
H05K 7/14	(2006.01)

(52) **U.S. Cl.**

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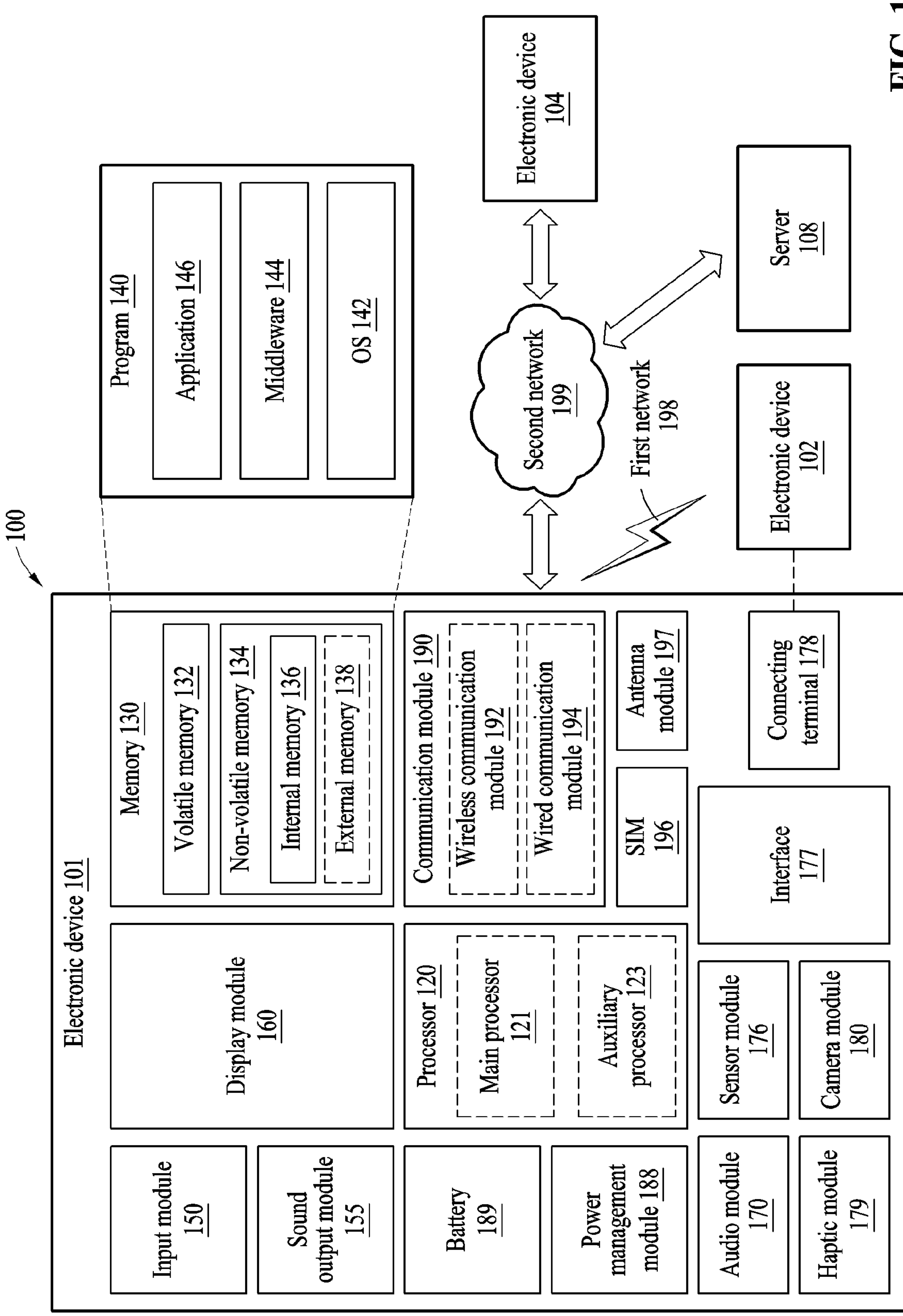
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Memory 130
Volatile memory 132
Non-volatile memory 134
Internal memory 136
External memory 138
Program 140
Application 146
Middleware 144
OS 142
Display module 160
Communication module 190
Wireless communication module 192
Wired communication module 194
SIM 196
Antenna module 197
Processor 120
Main processor 121
Auxiliary processor 123
Input module 150
Sound output module 155
Battery 189
Power management module 188
Audio module 170
Haptic module 179
Sensor module 176
Camera module 180
Interface 177
Connecting terminal 178
Second network 199
First network 198
Electronic device 104
Server 108
Electronic device 102

FIG. 1

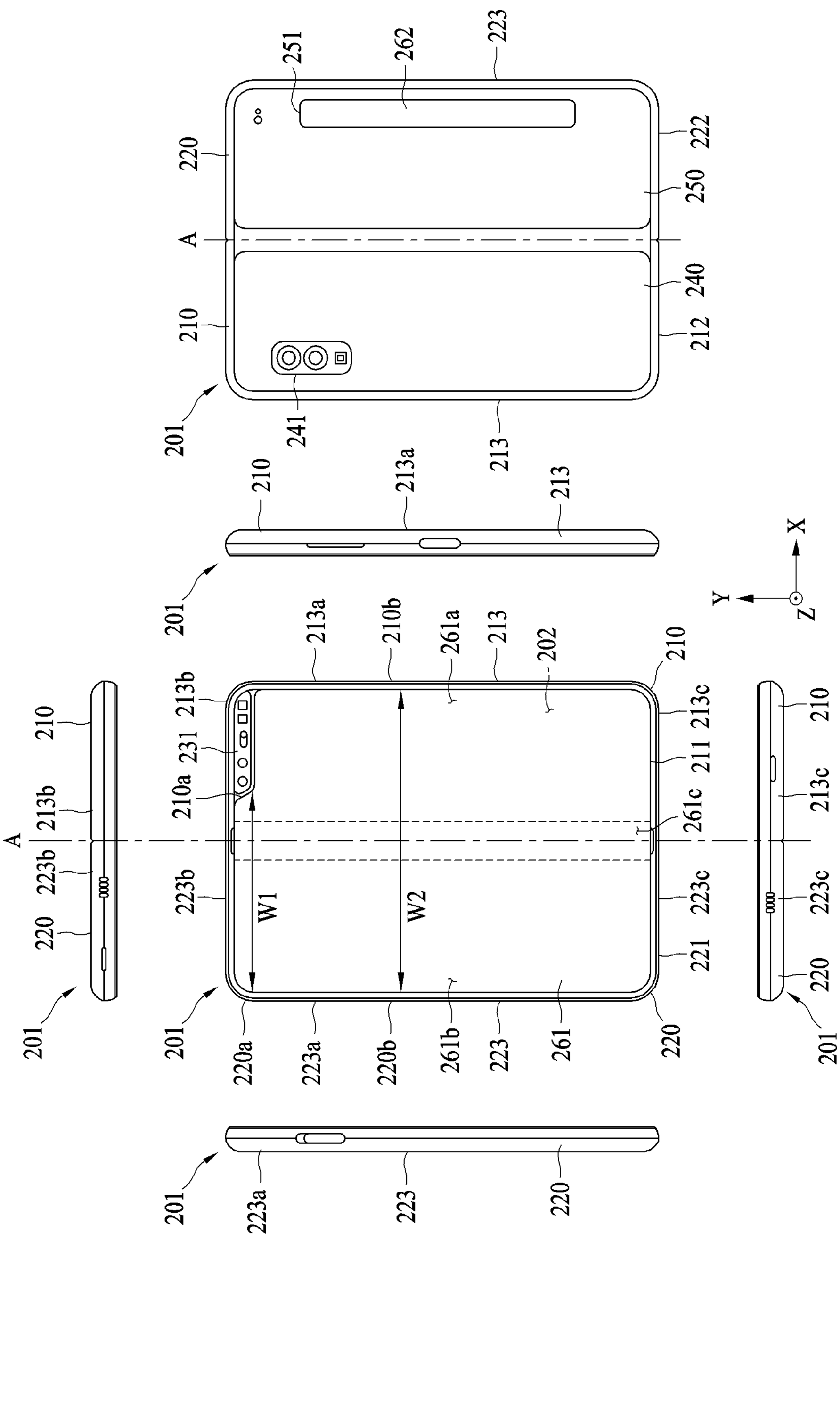
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FIG. 2A

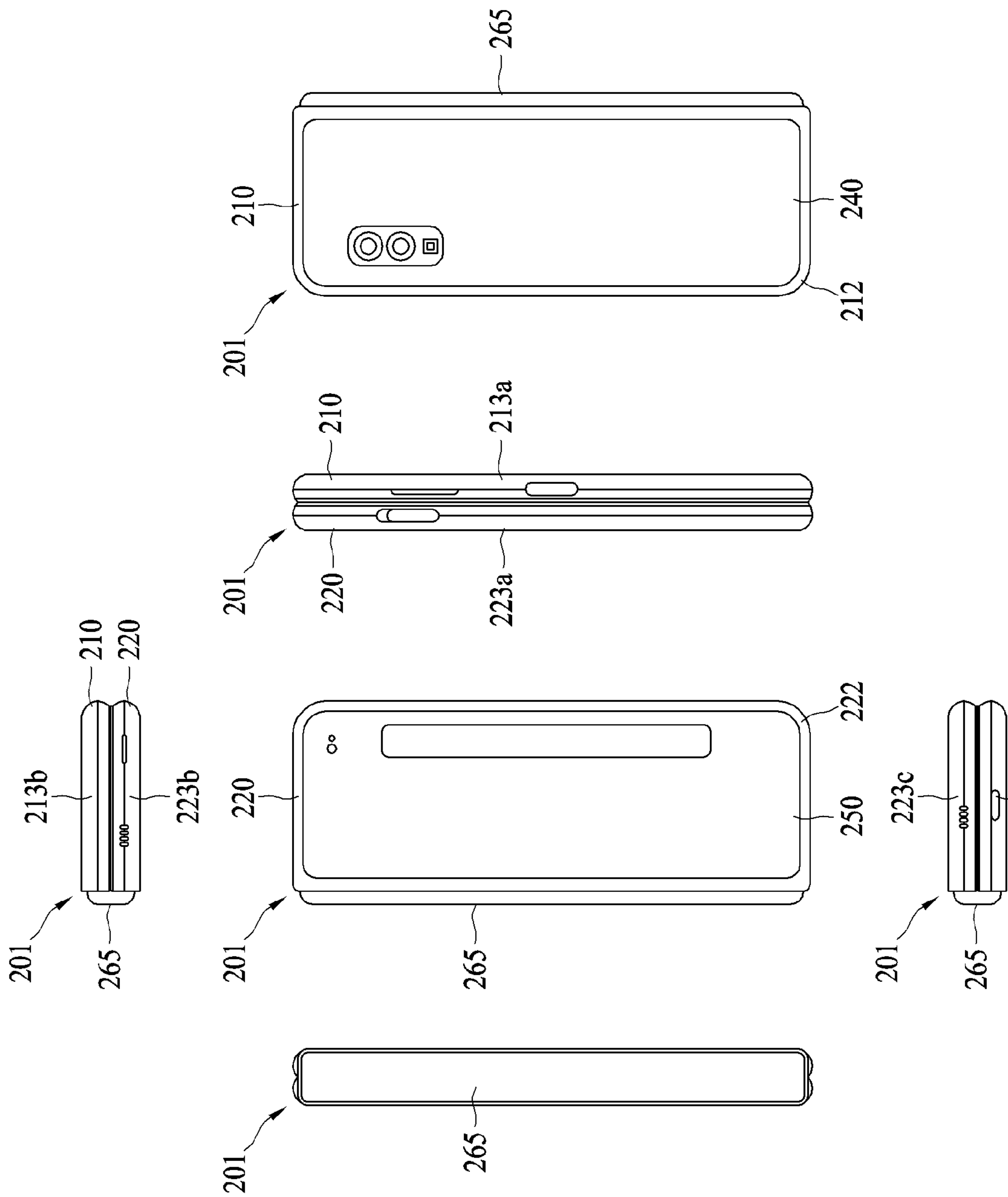
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FIG. 2B

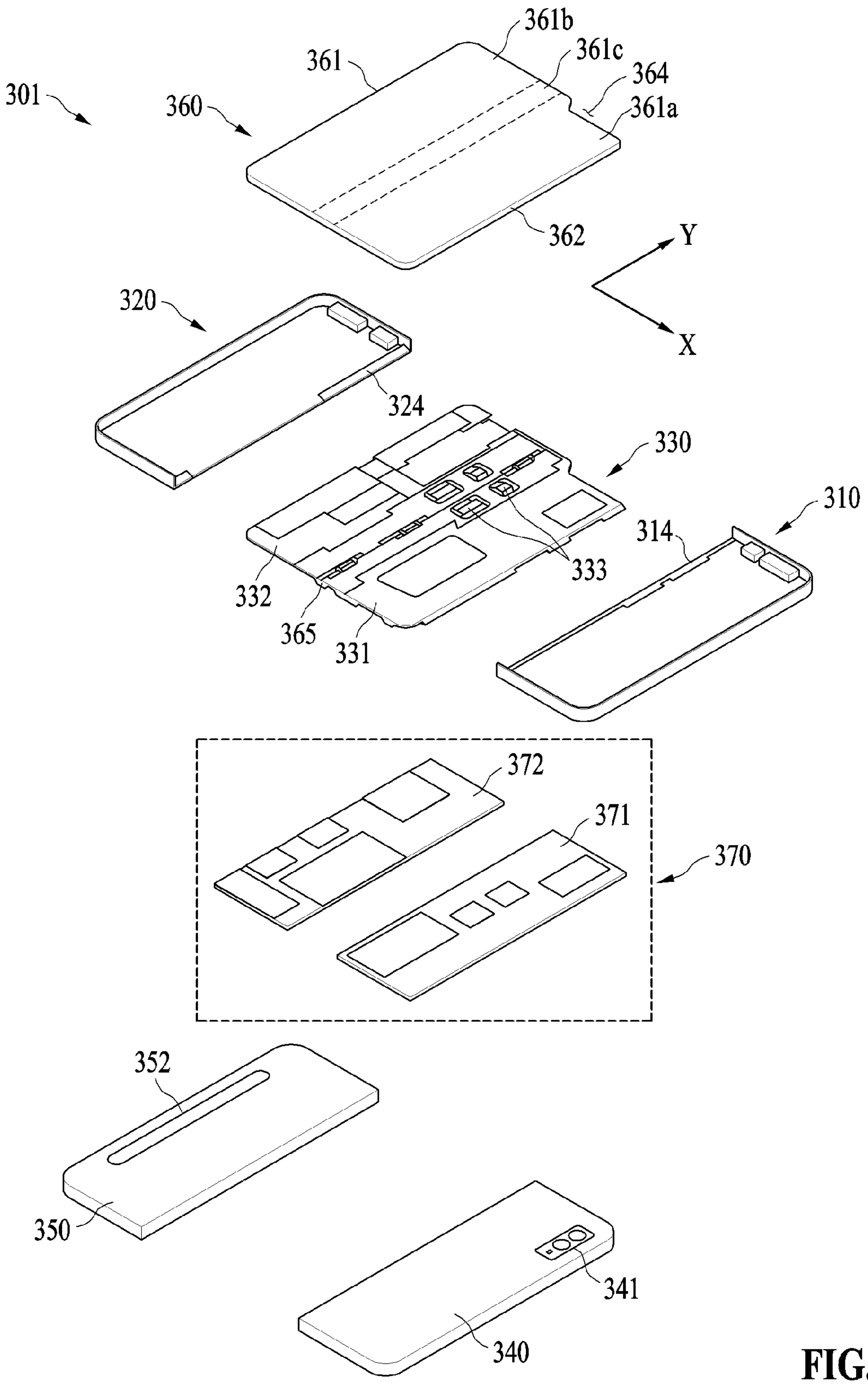

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FIG. 3

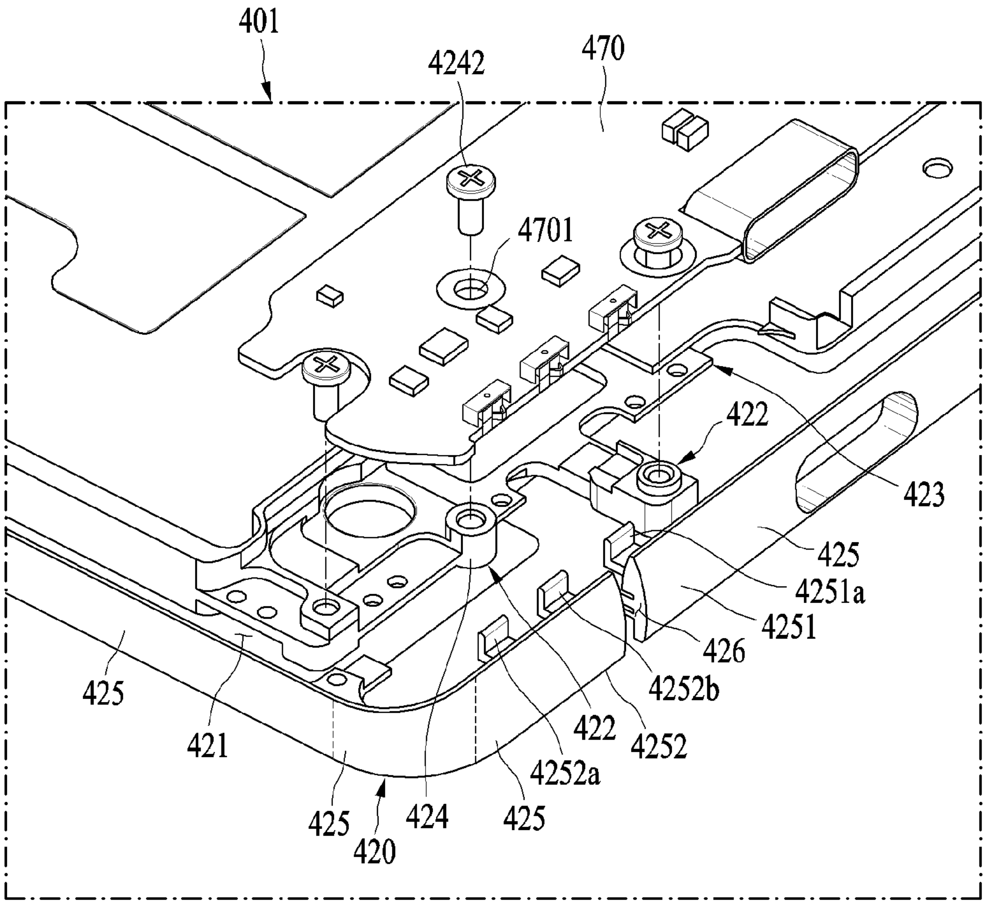
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FIG. 4A

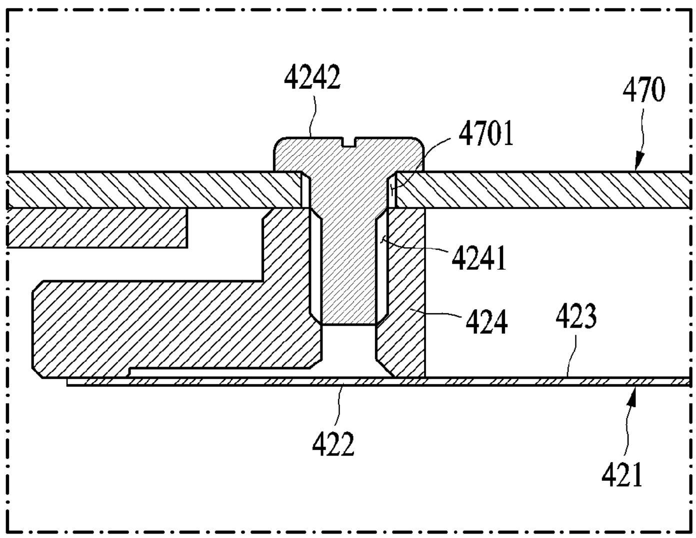

FIG. 4B

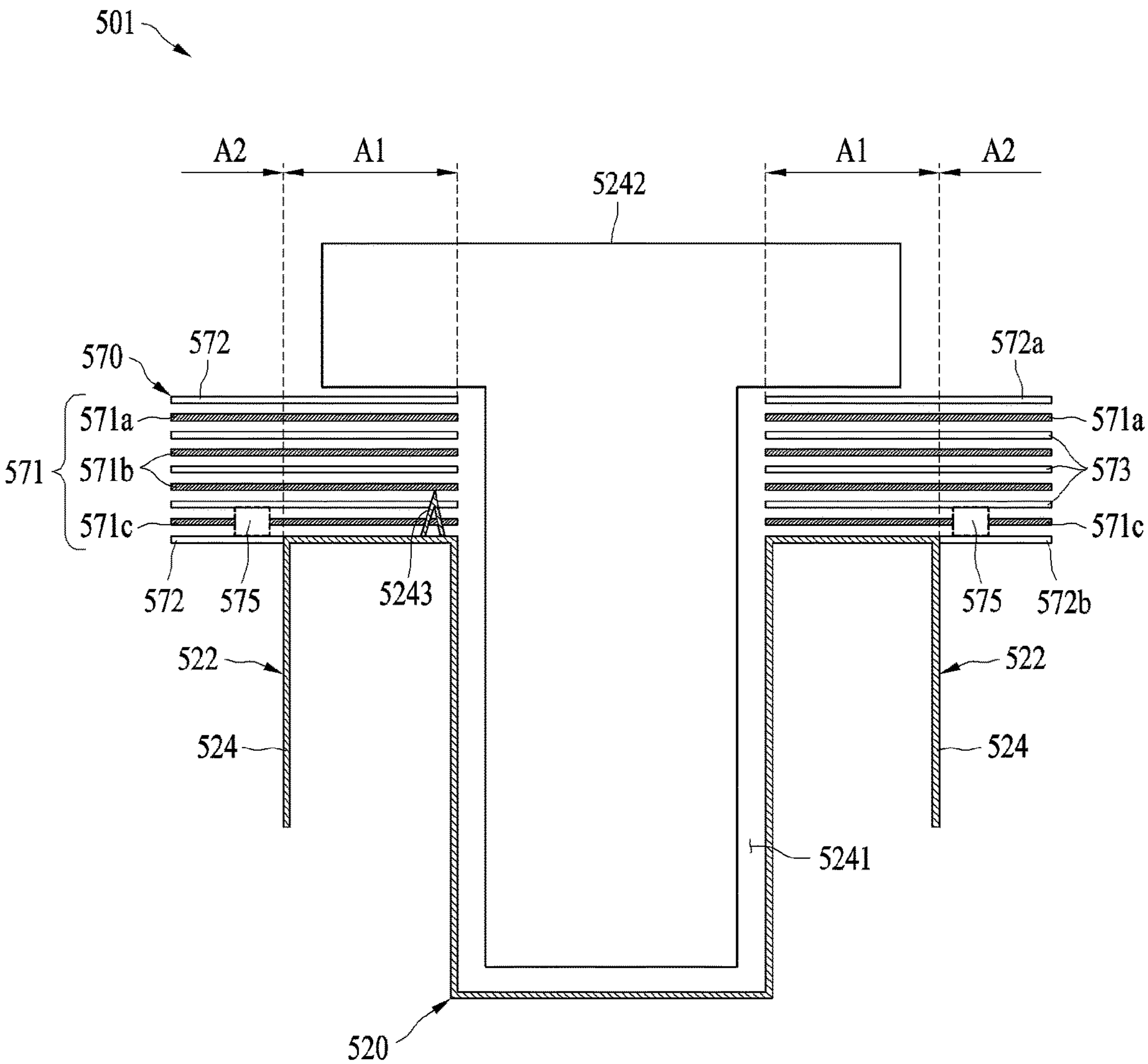
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FIG. 5A

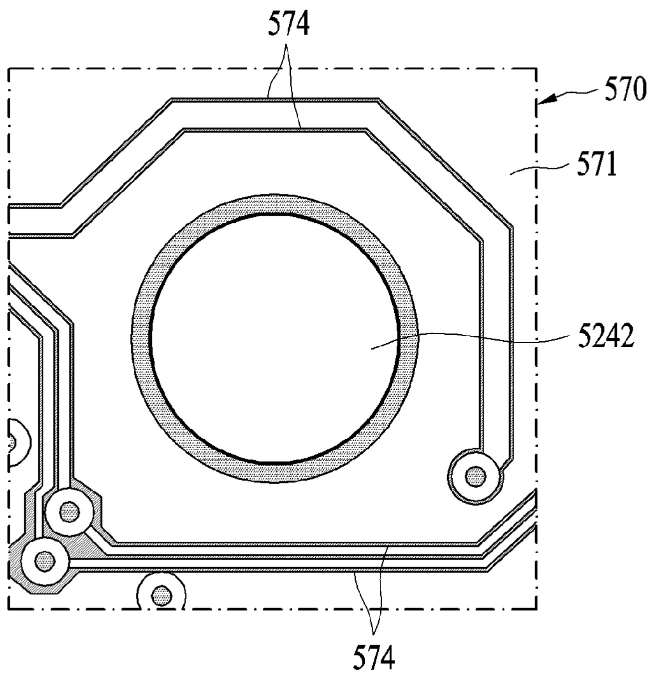

FIG. 5B

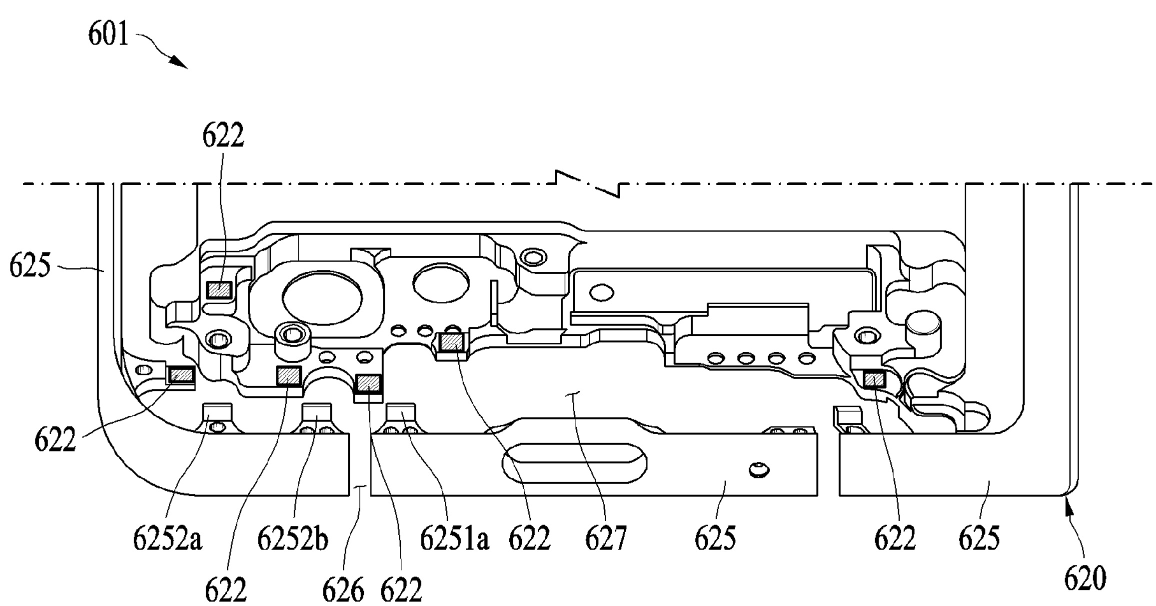
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FIG. 6A

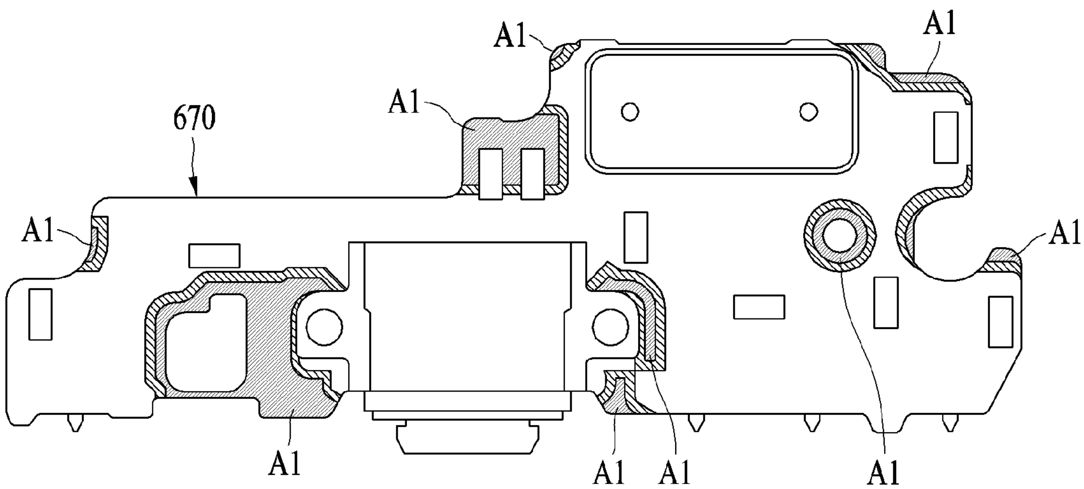


FIG. 6B

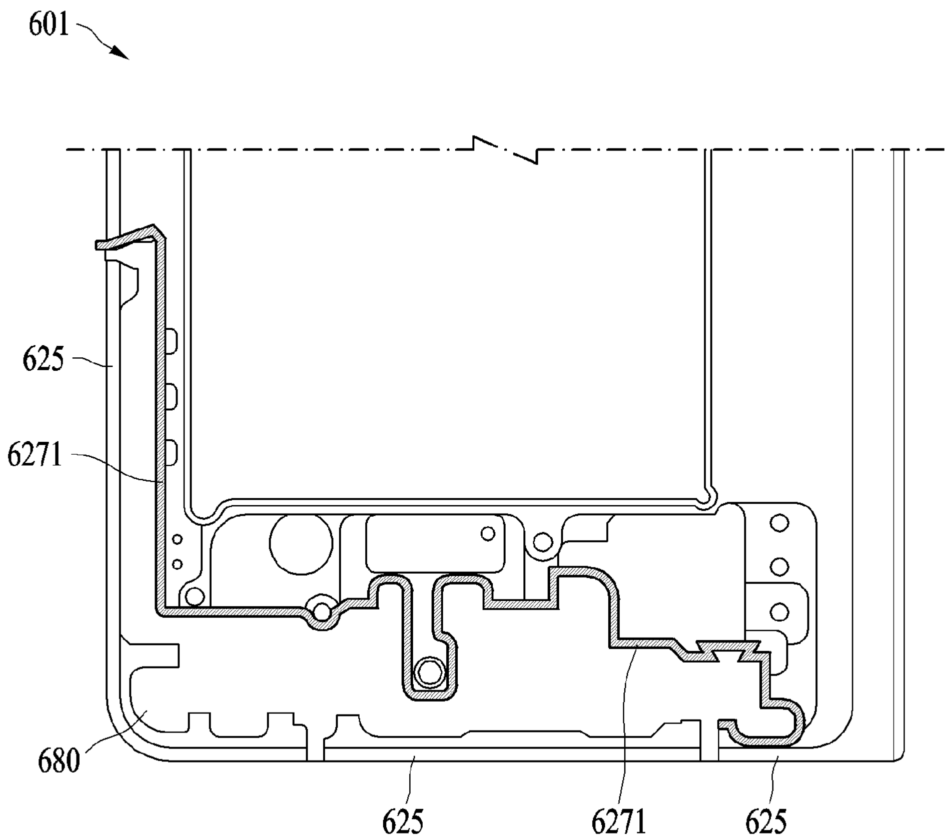
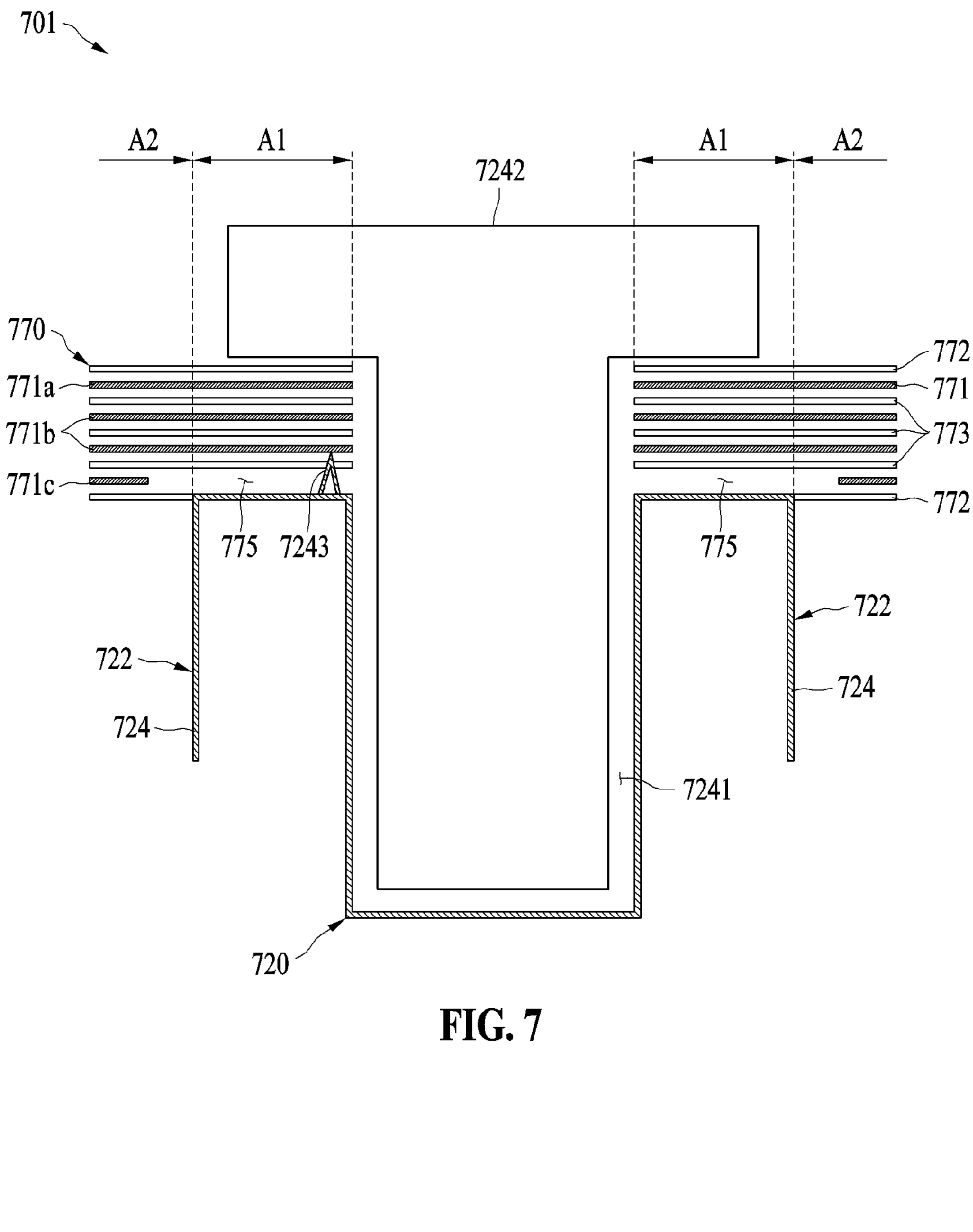


FIG. 6C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720

FIG. 7

ELECTRONIC APPARATUS FOR STABLE ELECTRICAL CONNECTION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of International Application No. PCT/KR2022/003839, filed on Mar. 18, 2022, which is based on and claims priority to Korean Patent Application No. 10-2021-0046941, filed on Apr. 12, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

The disclosure relates to an electronic device for a stable electrical connection.

2. Description of Related Art

Antennas used in electronic devices include monopole type antennas, and among the monopole type antennas, inverted F antennas or planar inverted F antennas are widely used. An antenna includes a feed portion and a ground portion, and the analysis and characteristics of a monopole antenna are on the premise that its ground portion is infinitely large and is manufactured as a perfect conductor. However, the ground portion is neither infinitely large nor a perfect conductor in practice. In consideration of components and structures in an electronic device, the ground portion may be formed in various geometrical shapes, and in some cases, a plurality of ground portions may function as one ground portion. In a case of a ground portion that is not sufficiently large, the ground portion may not perform its original function but perform as a radiator. Typically, in a case of a ground portion having a circular shape, the diameter of the ground portion may need to correspond to at least one wavelength of a signal. When the size of the ground portion is sufficiently large but is not manufactured as a conductor with desirable characteristics, all characteristics of the antenna may be affected thereby.

To connect a signal to an antenna in an electronic device, the electronic device may require a printed circuit board (PCB), and a ground portion of the PCB may function as a ground portion of the antenna, which may greatly affect the performance of the antenna. In addition, only the ground portion of the PCB having a small size may not implement a desired performance of the antenna, and thus a connection between the PCB and other electrical/mechanical component(s) in the electronic device may be considered important. Therefore, the ground portion of the PCB may need to be stably connected to other conductors to function as a wide ground portion, and structures including the ground portion of the PCB and its connection structure may need to be designed without deviation.

SUMMARY

Provided is an electronic device that implements a stable electrical connection between a printed circuit board (PCB) and peripheral conductive component(s).

According to an aspect of the disclosure, an electronic device includes: a housing including a coupling region; and a printed circuit board (PCB) including: an overlap region that overlaps the coupling region; a non-overlap region that does not overlap the coupling region; a plurality of metal layers including a plurality of lines; and a void portion in which a portion of at least one metal layer, among the plurality of metal layers, is not formed so that the at least one metal layer is discontinuous or terminated in the void portion, the at least one metal layer including a metal layer that is closest to the coupling region among the plurality of metal layers.

The at least one metal layer may not include a metal layer that is farthest from the coupling region among the plurality of metal layers.

The void portion may be in the non-overlap region, and the at least one metal layer may include only the metal layer that is closest to the coupling region.

The plurality of lines may be provided in both the overlap region and the non-overlap region.

The electronic device may further include an antenna including a ground portion, and the ground portion may be electrically connected to the metal layer that is closest to the coupling region among the plurality of metal layers.

The electronic device may further include an antenna, the housing may include an accommodating portion configured to accommodate the antenna, and the coupling region is at a boundary of the accommodating portion.

The plurality of lines may be lines through which signals pass.

The electronic device may further include: an antenna including a ground portion that is connected to the plurality of lines or forms at least a portion of the plurality of lines.

The housing may have a bottom surface including the coupling region and a non-coupling region that is different from the coupling region, and the housing may further include a protruding rib that protrudes from the coupling region of the bottom surface and contacts the PCB.

The housing may further include: a hole formed on the protruding rib; and a fixing member coupled to the hole and configured to fix the PCB to the protruding rib.

The protruding rib may include a burr contacting the metal layer that is closest to the coupling region.

The PCB may further include at least one dielectric layer on or under at least one metal layer among the plurality of metal layers.

The void portion may be in the non-overlap region.

The void portion may be in at least a portion of the overlap region and at least a portion of the non-overlap region.

The housing may further include a first housing including the coupling region and a second housing, and the electronic device may include: a hinge structure connecting the first housing and the second housing; and a display including a first area disposed in the first housing, a second area disposed in the second housing, and a flexible area between the first area and the second area.

According to one or more example embodiments, a stable electrical connection between a printed circuit board (PCB) and peripheral conductive component(s) may be implemented, and deviation of antenna performance in an electronic device may be reduced.

The effects of an electronic device according to various example embodiments are not limited to the effects described above, but other effects that are not described herein may be clearly understood from the following description by one of ordinary skill in the art.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects, features, and advantages of certain example embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

FIG. **1** is a block diagram illustrating an electronic device in a network environment according to one or more example embodiments;

FIG. **2**A is a view of an electronic device in an unfolded state according to one or more example embodiments;

FIG. **2**B is a view of an electronic device in a folded state according to one or more example embodiments;

FIG. **3** is an exploded perspective view of an electronic device according to one or more example embodiments;

FIG. **4**A is an exploded perspective view of a partial structure of an electronic device according to one or more embodiments;

FIG. **4**B is a view of a coupling structure between a housing and a printed circuit board (PCB) in the electronic device of FIG. **4**A;

FIG. **5**A is a view of a partial structure of an electronic device including a PCB according to one or more example embodiments;

FIG. **5**B is a view of a portion of a PCB including a plurality of lines according to one or more example embodiments;

FIG. **6**A is a view of regions coupled to a PCB among regions included in a housing of an electronic device according to one or more example embodiments.

FIG. **6**B is a view of regions coupled to coupling regions of the housing of FIG. **6**A among regions included in a PCB of an electronic device according to one or more example embodiments;

FIG. **6**C is a view of regions to which the structure of the PCB of FIG. **5** is applicable among regions included in a housing of an electronic device according to one or more example embodiments; and

FIG. **7** is a view of a partial structure of an electronic device including a PCB according to one or more example embodiments.

DETAILED DESCRIPTION

It is to be understood that various example embodiments of the disclosure and the terms used herein are not intended to limit the technological features set forth herein to particular example embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. In connection with the description of the drawings, like reference numerals may be used for similar or related components. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things unless the relevant context clearly indicates otherwise. As used herein, "A or B," "at least one of A and B," "at least one of A or B," "A, B, or C," "at least one of A, B, and C," and "at least one of A, B, or C," each of which may include any one of the items listed together in the corresponding one of the phrases, or all possible combinations thereof. Terms such as "1st" and "2nd" or "first" and "second" may simply be used to distinguish the component from other components in question, and do not limit the components in other aspects (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term "operatively" or "communicatively," as "coupled with," "coupled to," "connected with," or "connected to" another element (e.g., a second element), it denotes that the element may be coupled with the other element directly (e.g., by wire), wirelessly, or via a third element.

As used in connection with certain embodiments of the disclosure, the term "module" may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, "logic," "logic block," "part," or "circuitry." A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to one or more embodiments, the module may be implemented in the form of an application-specific integrated circuit (ASIC).

Various example embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., the internal memory **136** or the external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include code generated by a compiler or code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Here, the term "non-transitory" simply denotes that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

According to one or more embodiments, a method described herein may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., a compact disc read-only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smartphones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as a memory of the manufacturer's server, a server of the application store, or a relay server.

According to various example embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various example embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various example embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various example embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Referring to FIG. **1**, an electronic device **101** in a network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or communicate with at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). The electronic device **101** may communicate with the electronic device **104** via the server **108**. The electronic device **101** may include a processor **120**, a memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, and a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some example embodiments, at least one (e.g., the connecting terminal **178**) of the above components may be omitted from the electronic device **101**, or one or more other components may be added to the electronic device **101**. In some example embodiments, some (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) of the components may be integrated as a single component (e.g., the display module **160**).

The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** connected to the processor **120**, and may perform various data processing or computation. According to one or more example embodiments, as at least a part of data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**. The processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)) or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121** or to be specific to a specified function. The auxiliary processor **123** may be implemented separately from the main processor **121** or as a part of the main processor **121**.

The auxiliary processor **123** may control at least some of functions or states related to at least one (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) of the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state or along with the main processor **121** while the main processor **121** is an active state (e.g., executing an application). The auxiliary processor **123** (e.g., an ISP or a CP) may be implemented as a portion of another component (e.g., the camera module **180** or the communication module **190**) that is functionally related to the auxiliary processor **123**. The auxiliary processor **123** (e.g., an NPU) may include a hardware structure specifically for artificial intelligence (AI) model processing. An AI model may be generated by machine learning. The learning may be performed by, for example, the electronic device **101**, in which the AI model is performed, or performed via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, for example, supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The AI model may include a plurality of artificial neural network layers. An artificial neural network may include, for example, a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), and a bidirectional recurrent deep neural network (BRDNN), a deep Q-network, or a combination of two or more thereof, but is not limited thereto. The AI model may alternatively or additionally include a software structure other than the hardware structure.

The memory **130** may store various pieces of data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various pieces of data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

The program **140** may be stored as software in the memory **130** and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

The input module **150** may receive, from outside (e.g., a user) the electronic device **101**, a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

The sound output module **155** may output a sound signal to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing a recording. The receiver may be used to receive an incoming call. The receiver may be implemented separately from the speaker or as a part of the speaker.

The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector, and a control circuitry to control its corresponding one of the displays, the hologram device, and the projector. The display module **160** may include a touch sensor adapted to sense a touch, or a pressure sensor adapted to measure an intensity of a force of the touch.

The audio module **170** may convert sound into an electric signal or vice versa. The audio module **170** may obtain the sound via the input module **150** or output the sound via the sound output module **155** or an external electronic device (e.g., the electronic device **102**, such as a speaker or headphones) directly or wirelessly connected to the electronic device **101**.

The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101** and generate an electric signal or data value corresponding to the detected state. The sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

The interface **177** may support one or more specified protocols to be used by the electronic device **101** to couple with an external electronic device (e.g., the electronic device **102**) directly (e.g., by wire) or wirelessly. The interface **177** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

The connecting terminal **178** may include a connector via which the electronic device **101** may physically connect to an external electronic device (e.g., the electronic device **102**). The connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphones connector).

The haptic module **179** may convert an electric signal into a mechanical stimulus (e.g., a vibration or a movement) or an electrical stimulus, which may be recognized by a user via their tactile sensation or kinesthetic sensation. The haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

The camera module **180** may capture a still image and moving images. The camera module **180** may include one or more lenses, image sensors, ISPs, and flashes.

The power management module **188** may manage power supplied to the electronic device **101**. The power management module **188** may be implemented as, for example, at least a part of a power management integrated circuit (PMIC).

The battery **189** may supply power to at least one component of the electronic device **101**. The battery **189** may include, for example, a primary cell, which is not rechargeable, a secondary cell, which is rechargeable, or a fuel cell.

The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and an external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more CPs that are operable independently from the processor **120** (e.g., an AP) and that support direct (e.g., wired) communication or wireless communication. The communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device, for example, the electronic device **104**, via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5th generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., an LAN or a wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multiple components (e.g., multiple chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the SIM **196**.

The wireless communication module **192** may support a 5G network after a 4th generation (4G) network, and a next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., an mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (MIMO), full dimensional MIMO (FD-MIMO), an array antenna, analog beamforming, or a large-scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). The wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., an external electronic device) of the electronic device **101**. The antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). The antenna module **197** may include a plurality of antennas (e.g., an antenna array). In such a case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **198** or the second network **199**, may be selected by, for example, the communication module **190** from the plurality of antennas. The signal or power may be transmitted or received between the communication module **190** and the external electronic device via the at least one selected antenna. According to one or more example embodiments, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as a part of the antenna module **197**.

According to one or more example embodiments, the antenna module **197** may form an mmWave antenna module. The mmWave antenna module may include a PCB, an RFIC on a first surface (e.g., a bottom surface) of the PCB or adjacent to the first surface of the PCB and capable of supporting a designated high-frequency band (e.g., a mmWave band), and a plurality of antennas (e.g., an antenna array) disposed on a second surface (e.g., a top or a side surface) of the PCB, or adjacent to the second surface of the PCB and capable of transmitting or receiving signals in the designated high-frequency band.

At least some of the above-described components may be coupled mutually and exchange signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general-purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

According to one or more example embodiments, commands or data may be transmitted or received between the electronic device **101** and the external electronic device (e.g., the electronic device **104**) via the server **108** coupled with the second network **199**. Each of the external electronic devices (e.g., the electronic device **102** or **104**) may be a device of the same type as or a different type from the electronic device **101**. All or some of operations to be executed by the electronic device **101** may be executed by one or more of the external electronic devices (e.g., the electronic devices **102** and **104** and the server **108**). For example, if the electronic device **101** needs to perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or service, may request one or more external electronic devices to perform at least a part of the function or service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request and may transfer a result of the performance to the electronic device **101**. The electronic device **101** may provide the result, with or without further processing of the result, as at least part of a response to the request. To that end, cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra-low latency services using, e.g., distributed computing or MEC. According to another embodiment, the external electronic device (e.g., the electronic device **104**) may include an Internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. The external electronic device (e.g., the electronic device **104**) or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., a smart home, a smart city, a smart car, or healthcare) based on 5G communication technology or IoT-related technology.

According to various example embodiments described herein, an electronic device may be a device of any one of various types. The electronic device may include, as non-limiting examples, a portable communication device (e.g., a smartphone), a computing device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. However, the electronic device is not limited to the foregoing examples.

Referring to FIGS. **2**A and **2**B, a foldable electronic device **201** may include a pair of housings **210** and **220** rotatably coupled to each other through a hinge structure to be folded with respect to each other, a hinge cover **265** for covering foldable portions of the pair of housings **210** and **220**, and a display **261** (e.g., a flexible display or a foldable display) disposed in a space formed by the pair of housings **210** and **220**. In one or more example embodiments, a surface on which the display **261** is disposed may be defined as a front surface of the foldable electronic device **201**, and a surface opposite to the front surface may be defined as a rear surface of the foldable electronic device **201**. In addition, a surface surrounding a space between the front surface and the rear surface may be defined as a side surface of the foldable electronic device **201**.

In one or more example embodiments, the pair of housings **210** and **220** may include a first housing **210**, a second housing **220**, a first rear cover **240**, and a second rear cover **250**. The pair of housings **210** and **220** of the electronic device **201** are not limited to the shapes or the combination and/or coupling of components shown in FIGS. **2**A and **2**B but may be implemented in other shapes or by another combination and/or coupling of components.

In one or more example embodiments, the first housing **210** and the second housing **220** may be disposed on both sides with respect to a folding axis A and may be substantially symmetrical to the folding axis A. In one or more example embodiments, an angle or distance between the first housing **210** and the second housing **220** may vary depending on whether the electronic device **201** is in an unfolded state, a folded state, or an intermediate state. In one or more example embodiments, unlike the second housing **220**, the first housing **210** may include a sensor area **231** in which various sensor modules (e.g., the sensor module **176** of FIG. **1**) are disposed, and the first housing **210** and the second housing **220** may have shapes symmetrical to each other in areas other than the sensor area **231**. In one or more other example embodiments, the sensor area **231** may be disposed at least a partial area of the second housing **220**. In one or more other example embodiments, the sensor area **231** may be replaced with at least a partial area of the second housing **220**. The sensor area **231** may include, for example, a camera hole area, a sensor hole area, an under-display camera (UDC) area, and/or an under-display sensor (UDS) area.

In one or more example embodiments, the first housing **210** may be connected to the hinge structure in the unfolded state of the electronic device **201**. The first housing **210** may include a first surface **211** facing the front surface of the electronic device **201**, a second surface **212** facing a direction opposite to the first surface **211**, and a first side portion **213** enclosing at least a portion of a space between the first surface **211** and the second surface **212**. The first side portion **213** may include a first side surface **213***a* disposed substantially in parallel to the folding axis A, a second side surface **213***b* extending in a direction substantially perpendicular to the folding axis A from one end of the first side surface **213***a*, and a third side surface **213***c* extending in a direction substantially perpendicular to the folding axis A from another end of the first side surface **213***a* and substantially parallel to the second side surface **213***b*. The second housing **220** may be connected to the hinge structure in the unfolded state of the electronic device **201**. The second housing **220** may include a third surface **221** facing the front surface of the electronic device **201**, a fourth surface **222** facing a direction opposite to the third surface **221**, and a second side portion **223** enclosing at least a portion of a space between the third surface **221** and the fourth surface **222**. The second side portion **223** may include a fourth side surface **223***a* disposed substantially in parallel to the folding axis A, a fifth side surface **223***b* extending in a direction substantially perpendicular to the folding axis A from one end of the fourth side surface **223***a*, and a sixth side surface **223***c* extending in a direction substantially perpendicular to the folding axis A from another end of the fourth side surface **223***a* and substantially parallel to the fifth side surface **223***b*. The first surface **211** and the third surface **221** may face each other when the electronic device **201** is in the folded state.

In one or more example embodiments, the electronic device **201** may include a recessed accommodating portion **202** for accommodating the display **261** through the structural coupling of the first housing **210** and the second housing **220**. The accommodating portion **202** may have substantially the same size as the display **261**. In one or more example embodiments, due to the sensor area **231**, the accommodating portion **202** may have at least two different widths in a direction perpendicular to the folding axis A. For example, the accommodating portion **202** may have a first width W**1** between a first portion **210***a* formed at an edge of the sensor area **231** of the first housing **210** and a second portion **220***a* parallel to the folding axis A of the second housing **220**, and a second width W**2** between a third portion **210***b* parallel to the folding axis A without overlapping the sensor area **231** of the first housing **210** and a fourth portion **220***b* of the second housing **220**. The second width W**2** may be greater than the first width W**1**. That is, the accommodating portion **202** may be formed to have the first width W**1** from the first portion **210***a* of the first housing **210** to the second portion **220***a* of the second housing **220** and the second width W2 from the third portion **210***b* of the first housing **210** to the fourth portion **220***b* of the second housing **220**, which have a mutually asymmetrical shape. The first portion **210***a* and the third portion **210***b* of the first housing **210** may be formed at different distances from the folding axis A. However, the width of the accommodating portion **202** may not be limited to the illustrated example. For example, the accommodating portion **202** may have three or more different widths by the shape of the sensor area **231** or the asymmetrical shape of the first housing **210** and the second housing **220**.

In one or more example embodiments, at least a portion of the first housing **210** and the second housing **220** may be formed of a metal material or a non-metal material having a predetermined magnitude of rigidity appropriate to support the display **261**.

In one or more example embodiments, the sensor area **231** may be formed adjacent to one corner of the first housing **210**. However, the arrangement, shape, and size of the sensor area **231** are not limited to the illustrated example. In one or more other example embodiments, the sensor area **231** may be formed at another corner of the first housing **210** or in a predetermined area of an upper corner and a lower corner. In one or more other example embodiments, the sensor area **231** may be disposed in at least a partial area of the second housing **220**. In one or more example embodiments, the sensor area **231** may be formed in a shape extending between the first housing **210** and the second housing **220**.

In one or more example embodiments, the electronic device **201** may include at least one component disposed to be exposed on the front surface of the electronic device **201** through the sensor area **231** or at least one opening formed in the sensor area **231**. The component may include, for example, at least one of a front camera module, a receiver, a proximity sensor, an illuminance sensor, an iris recognition sensor, an ultrasonic sensor, or an indicator.

In one or more example embodiments, the first rear cover **240** may be disposed on the second surface **212** of the first housing **210** and may have substantially rectangular edges. At least a portion of the edges of the first rear cover **240** may be surrounded by the first housing **210**. The second rear cover **250** may be disposed on the fourth surface **222** of the second housing **220** and may have substantially rectangular edges. At least a portion of the edges of the second rear cover **250** may be surrounded by the second housing **220**.

In one or more example embodiments, the first rear cover **240** and the second rear cover **250** may have substantially symmetrical shapes with respect to the folding axis A. In one or more other example embodiments, the first rear cover **240** and the second rear cover **250** may have different shapes. In still one or more other example embodiments, the first housing **210** and the first rear cover **240** may be integrally formed, and the second housing **220** and the second rear cover **250** may be integrally formed.

In one or more example embodiments, the first housing **210**, the second housing **220**, the first rear cover **240**, and the second rear cover **250** may provide a space in which various components (e.g., a PCB, the antenna module **197** of FIG. **1**, the sensor module **176** of FIG. **1**, or the battery **189** of FIG. **1**) of the electronic device **201** may be arranged through a structure in which the first housing **210**, the second housing **220**, the first rear cover **240**, and the second rear cover **250** are coupled to one another. In one or more example embodiments, at least one component may be visually exposed on the rear surface of the electronic device **201**. For example, at least one component may be visually exposed through a first rear area **241** of the first rear cover **240**. The component may include, for example, a proximity sensor, a rear camera module, and/or a flash. In one or more example embodiments, at least a portion of a sub-display **262** may be visually exposed through a second rear area **251** of the second rear cover **250**. In one or more example embodiments, the electronic device **201** may include a sound output module (e.g., the sound output module **155** of FIG. **1**) disposed through at least a partial area of the second rear cover **250**.

In one or more example embodiments, the display **261** may be disposed in the accommodating portion **202** formed by the pair of housings **210** and **220**. For example, the display **261** may be formed to occupy substantially most of the front surface of the electronic device **201**. The front surface of the electronic device **201** may include an area in which the display **261** is disposed, and a partial area (e.g., an edge area) of the first housing **210** and a partial area (e.g., an edge area) of the second housing **220**, which are adjacent to the display **261**. The rear surface of the electronic device **201** may include the first rear cover **240**, a partial area (e.g., an edge area) of the first housing **210** adjacent to the first rear cover **240**, the second rear cover **250**, and a partial area (e.g., an edge area) of the second housing **220** adjacent to the second rear cover **250**. In one or more example embodiments, the display **261** may be a display in which at least one area is deformable into a planar surface or a curved surface. In one or more example embodiments, the display **261** may include a folding area **261***c*, a first area **261***a* on a first side (e.g., the right side) of the folding area **261***c*, and a second area **261***b* on a second side (e.g., the left side) of the folding area **261***c*. For example, the first area **261***a* may be disposed on the first surface **211** of the first housing **210**, and the second area **261***b* may be disposed on the third surface **221** of the second housing **220**. However, the area division of the display **261** is provided merely as an example, and the display **261** may be divided into a plurality of areas depending on the structure or functions of the display **261**. For example, as shown in FIG. **2**A, the display **261** may be divided into areas based on the folding axis A or the folding area **261***c* extending in parallel to a Y-axis, or the display **261** may be divided into areas based on another folding area (e.g., a folding area extending in parallel to an X-axis) or another folding axis (e.g., a folding axis parallel to the X-axis). This area division of the display **261** is provided merely as an example of physical division based on the pair of housings **210** and **220** and the hinge structure, and the display **261** may display substantially one screen through the pair of housings **210** and **220** and the hinge structure. In one or more example embodiments, the first area **261***a* may include a notch area formed along the sensor area **231**, but the other areas of the first area **261***a* may be substantially symmetrical to the second area **261***b*. In one or more other example embodiments, the first area **261***a* and the second area **261***b* may have substantially symmetrical shapes with respect to the folding area **261***c*.

In one or more example embodiments, the hinge cover **265** may be disposed between the first housing **210** and the second housing **220** and configured to cover the hinge structure. The hinge cover **265** may be hidden by at least a portion of the first housing **210** and the second housing **220** or exposed to the outside according to an operating state of the electronic device **201**. For example, when the electronic device **201** is in an unfolded state as shown in FIG. **2**A, the hinge cover **265** may be hidden by the first housing **210** and the second housing **220** so that it is not exposed to the outside, and when the electronic device **201** is in a folded state as shown in FIG. **2**B, the hinge cover **265** may be exposed to the outside between the first housing **210** and the second housing **22**. In addition, when the electronic device **201** is in an intermediate state in which the first housing **210** and the second housing **220** form an angle with each other, at least a portion of the hinge cover **265** may be exposed to the outside between the first housing **210** and the second housing **220**. In this case, an area of the hinge cover **265** exposed to the outside may be smaller than an area of the hinge cover **265** exposed when the electronic device **201** is in the folded state. In one or more example embodiments, the hinge cover **265** may have a curved surface.

The operations of the electronic device **201** will be described hereinafter. When the electronic device **201** is in the unfolded state (e.g., a state of the electronic device **201** of FIG. **2**A), the first housing **210** and the second housing **220** may form a first angle (e.g., approximately 180 degrees (°)) with each other, and the first area **261***a* and the second area **261***b* of the display **261** may be oriented in substantially the same direction. The folding area **261***c* of the display **261** may be on substantially the same plane as the first area **261***a* and the second area **261***b*. In one or more other example embodiments, when the electronic device **201** is in the unfolded state, the first housing **210** may rotate at a second angle (e.g., approximately 360°) relative to the second housing **220**, whereby the first housing **210** and the second housing **220** may be reversely folded such that the second surface **212** and the fourth surface **222** may face each other.

In addition, when the electronic device **201** is in the folded state (e.g., a state of the electronic device **201** of FIG. **2**B), the first housing **210** and the second housing **220** may face each other. The first housing **210** and the second housing **220** may form an angle of approximately 0° to approximately 10°, and the first area **261***a* and the second area **261***b* of the display **261** may face each other. At least a portion of the folding area **261***c* of the display **261** may be deformed into a curved surface.

In addition, when the electronic device **201** is in the intermediate state, the first housing **210** and the second housing **220** may form a predetermined angle with each other. An angle (e.g., a third angle, approximately 90°) formed by the first area **261***a* and the second area **261***b* of the display **261** may be greater than an angle formed when the electronic device **201** is in the folded state and less than an angle formed when the electronic device **201** is in the unfolded state. At least a portion of the folding area **261***c* of the display **261** may be deformed into a curved surface. In this case, a curvature of the curved surface of the folding area **261***c* may be smaller than that formed when the electronic device **201** is in the folded state.

Various example embodiments of the electronic device described herein are not limited to a form factor of the electronic device **201** described with reference to FIGS. **2**A and **2**B, and may also apply to electronic devices with various form factors.

Referring to FIG. **3**, according to various example embodiments, an electronic device **301** may include a display module **360** (e.g., the display module **160**), a hinge assembly **330**, a substrate **370**, a first housing **310** (e.g., the first housing **210**), a second housing **320** (e.g., the second housing **220**), a first rear cover **340** (e.g., the first rear cover **240**) including a first rear area **341** (e.g., the first rear area **241**), and a second rear cover **350** (e.g., the second rear cover **250**) including a second rear area **351** (e.g., the second rear area **251**).

The display module **360** may include a display **361** (e.g., the display **261**) and at least one layer or plate **362** on which the display **361** is disposed. In one or more example embodiments, the plate **362** may be disposed between display **361** and a hinge assembly **330**. The display **361** may be disposed on at least a portion of one surface (e.g., an upper surface) of the plate **362**. The plate **362** may be formed in a shape corresponding to that of the display **361**. For example, a partial area of the plate **362** may be formed in a shape corresponding to a notch area **364** of the display **361**.

The hinge assembly **330** may include a first bracket **331**, a second bracket **332**, a hinge structure disposed between the first bracket **331** and the second bracket **332**, a hinge cover **365** configured to cover the hinge structure when viewed from the outside, and a wiring member **333** (e.g., a flexible printed circuit (FPC)) crossing the first bracket **331** and the second bracket **332**.

In one or more example embodiments, the hinge assembly **330** may be disposed between the plate **362** and the substrate **370**. For example, the first bracket **331** may be disposed between the first area **361***a* of the display **361** and a first substrate **371**. The second bracket **332** may be disposed between the second area **361***b* of the display **361** and a second substrate **372**.

In one or more example embodiments, at least a portion of the wiring member **333** and the hinge structure may be disposed inside the hinge assembly **330**. The wiring member **333** may be disposed in a direction (e.g., an X-axis direction) crossing the first bracket **331** and the second bracket **332**. The wiring member **333** may be disposed in a direction (e.g., the x-axis direction) substantially perpendicular to a folding axis (e.g., the Y-axis or the folding axis A of FIG. **2**A) of a flexible area **361***c* of the electronic device **301**.

The substrate **370** may include the first substrate **371** disposed on the side of the first bracket **331** and the second substrate **372** disposed on the side of the second bracket **332**. The first substrate **371** and the second substrate **372** may be arranged inside a space formed by the hinge assembly **330**, the first housing **310**, the second housing **320**, the first rear cover **340**, and the second rear cover **350**. On the first substrate **371** and the second substrate **372**, components for implementing various functions of the electronic device **301** may be mounted.

The first housing **310** and the second housing **320** may be assembled to each other so as to be coupled to both sides of the hinge assembly **330** while the display module **360** is coupled to the hinge assembly **330**. The first housing **310** and the second housing **320** may be coupled to the hinge assembly **330** by sliding on both sides of the hinge assembly **330**.

In one or more example embodiments, the first housing **310** may include a first rotation support surface **314**, and the second housing **320** may include a second rotation support surface **324** corresponding to the first rotation support surface **314**. The first rotation support surface **314** and the second rotation support surface **324** may each include a curved surface corresponding to a curved surface included in the hinge cover **365**.

In one or more example embodiments, when the electronic device **301** is in an unfolded state (e.g., a state of the electronic device **201** of FIG. **2**A), the first rotation support surface **314** and the second rotation support surface **324** may cover the hinge cover **365** such that the hinge cover **365** may not be exposed through the rear surface of the electronic device **301** or may be minimally exposed. In addition, when the electronic device **301** is in a folded state (e.g., a state of the electronic device **201** of FIG. **2**B), the first rotation support surface **314** and the second rotation support surface **324** may rotate along the curved surface included in the hinge cover **365** such that the hinge cover **365** may be maximally exposed through the rear surface of the electronic device **301**.

Referring to FIGS. **4**A and **4**B, according to one or more example embodiments, an electronic device **401** (e.g., the electronic device **301**) may include at least one housing **420** (e.g., the first housing **310** and/or the second housing **320**), and the at least one housing **420** may include a plurality of side members **425** that forms a side frame defining an internal space of the housing **420**. In one or more example embodiments, a slit **426** may be formed between any pair of neighboring side members **425** among the plurality of side members **425**. In one or more example embodiments, some side members **425** among the plurality of side members **425** may have a round shape.

In one or more example embodiments, at least one first feed terminal **4251***a* may be disposed on a first side member **4251** of the plurality of side members **425**, and at least one second feed terminal **4252***a* and/or at least one ground terminal **4252***b* may be disposed on a second side member **4252** of the plurality of side members **425**. In one or more example embodiments, the plurality of side members **425** may at least partially include a conductive material.

In one or more example embodiments, at least some of the plurality of side members **425** may serve as a radiating conductor of an antenna. For example, a processor (e.g., the processor **120**) and/or a communication module (e.g., the communication module **190**) of the electronic device **401** may perform wireless communication through one or more of the plurality of side members **425** that serves as the radiating conductor of the antenna.

In one or more example embodiments, one or more of the plurality of side members **425** may be electrically connected to a PCB **470**. For example, the at least one first feed terminal **4251***a* disposed on the first side member **4251** and/or the at least one second feed terminal **4252***a* disposed on the second side member **4252** may be electrically connected to one portion of the PCB **470**, the at least one ground terminal **4252***b* disposed on the second side member **4252** may be electrically connected to another portion of the PCB **470**, and a conductive material portion forming the at least some of the side members **425** may form a portion of the antenna between a portion in which the at least one first feed terminal **4251***a* is disposed and/or the at least one second feed terminal **4252***a* is disposed and a portion in which the at least one ground terminal **4252***b* is disposed.

In one or more example embodiments, the housing **420** may include a bottom surface **421** disposed in the internal space formed by the plurality of side members **425**, and the bottom surface **421** may include a coupling region **422** coupled to the PCB **470** and a non-coupling region **423** that is not coupled to the PCB **470**. In one or more example embodiments, the housing **420** may include a protruding rib **424** protruding from the coupling region **422**. One surface (e.g., an upper surface) of the protruding rib **424** may contact a partial area (e.g., an overlap region A**1**) of the PCB **470** and support the PCB **470**. In one or more example embodiments, the protruding rib **424** may be fastened to the PCB **470**. For example, the PCB **470** and the protruding rib **424** may include holes **4701** and **4241**, respectively, and a fixing member **4242** may fix the PCB **470** to the protruding rib **424** through the hole **4701** of the PCB **470** and the hole **4241** of the protruding rib **424**. In one or more example embodiments, the fixing member **4242** may include a structure, such as, for example, a screw, a hook, and/or other fixing structures. The shape and structure of the coupling region **422** that is to be coupled to the PCB **470** are not limited to the shape and structure of the protruding rib **424** described above, and there may be components of various shapes and structures.

Referring to FIGS. **5**A and **5**B, according to one or more example embodiments, an electronic device **501** (e.g., the electronic device **401**) may include a housing **520** (e.g., the housing **420**) and a PCB **570** (e.g., the PCB **470**) coupled to the housing **520**.

In one or more example embodiments, the housing **520** may include a protruding rib **524** (e.g., the protruding rib **424**) protruding from a coupling region **522** (e.g., the coupling region **422**) that is coupled to the PCB **570**, a hole **5241** (e.g., the hole **4241**) formed in the protruding rib **524**, and a fixing member **5242** (e.g., the fixing member **4242**) that fixes the PCB **570** to the protruding rib **524** and is fastened to the hole **5241**.

In one or more example embodiments, in a case in which at least a portion of the coupling region **522** is formed of a metal material, the coupling region **522** may include a burr **5243**. For example, the burr **5243** may be formed on one surface (e.g., the upper surface) of the protruding rib **524** in contact with the PCB **570**. The term "burr" used herein indicates an end curl of a metal cut part.

In one or more example embodiments, the PCB **570** may include a plurality of metal layers **571**, a plurality of printed solder resists **572**, and a plurality of dielectrics **573**. The plurality of metal layers **571** may include, for example, a first metal layer **571***a* (e.g., an upper layer), at least one second metal layer **571***b* (e.g., a middle layer), and a third metal layer **571***c* (e.g., a lower layer). The plurality of metal layers **571**, the plurality of printed solder resists **572**, and the plurality of dielectrics **573** may be stacked in a thickness direction of the PCB **570**. In one or more example embodiments, a first printed solder resist **572***a* of the plurality of printed solder resists **572** may be disposed on the first metal layer **571***a* of the plurality of metal layers **571**, and a second printed solder resist **572***b* of the plurality of printed solder resists **572** may be disposed under the third metal layer **571***c* of the plurality of metal layers **571**. Each of the plurality of dielectrics **573** may be disposed between the first metal layer **571***a* and the at least one second metal layer **571***b* and/or between the at least one second metal layer **571***b* and the third metal layer **571***c*. When there is a plurality of second metal layers **571***b*, at least some of the plurality of dielectrics **573** may be disposed between a pair of neighboring second metal layers **571***b*. Such an arrangement of the plurality of metal layers **571**, the plurality of printed solder resists **572**, and the plurality of dielectrics **573** is not limited to the illustrated example, and various arrangements may also be applicable.

In one or more example embodiments, one or more of the plurality of metal layers **571** may be electrically connected to a ground portion (e.g., the ground terminal **4252***b*) of an antenna of the electronic device **501**. For example, the at least one second metal layer **571***b* and the third metal layer **571***c* may be connected to the ground portion of the antenna of the electronic device **501**. For another example, only the third metal layer **571***c* may be connected to the ground portion of the antenna of the electronic device **501**.

In one or more example embodiments, the PCB **570** may include an overlap region **A1** that overlaps the coupling region **522** of the housing **520** and a non-overlap region A**2** that does not overlap the coupling region **522**. For example, the overlap region **A1** may indicate a region that overlaps at least a portion of the protruding rib **524**, while the non-overlap region **A2** may indicate a region that does not overlap any portion of the protruding rib **524**.

In one or more example embodiments, the plurality of metal layers **571** may include a plurality of lines **574**. For example, the plurality of lines **574** may be signal lines through which electrical signals pass. For another example, the plurality of lines **574** may be a portion of the ground portion (e.g., the ground terminal **4252***b*) that forms a portion of the antenna of the electronic device **501**. For still another example, the plurality of lines **574** may be electrically connected to the ground portion that forms a portion of the antenna of the electronic device **501**.

In one or more example embodiments, the plurality of lines **574** may be formed throughout both the overlap region A**1** and the non-overlap region A**2** of the PCB **570**. That is, the plurality of lines **574** may be formed by maximally utilizing substantially most of the limited areas of the PCB **570** and may thus maximize the wiring efficiency of the PCB **570**.

In one or more example embodiments, the PCB **570** may include a void portion **575** that is formed because a portion of at least one metal layer (e.g., the at least one second metal layer **571***b* and/or the third metal layer **571***c*) close to the coupling region **522** (e.g., one surface of the protruding rib **524**) among the plurality of metal layers **571** is not formed, in the non-overlap region A**2**. That is, at the void portion **575**, the at least one metal layer (e.g., the at least one second metal layer **571** and/or the third metal layer **571***c*) close to the coupling region **522** (e.g., one surface of the protruding rib **524**), among the plurality of metal layers **571**, may be electrically opened or discontinuous. When the burr **5243** that may occur in the coupling region **522** is in contact with at least one metal layer in which the void portion **575** is formed, this may prevent a direct contact between or an electrical connection between the burr **5243** and a line formed on the at least one metal layer and may thereby ensure a stable electrical connection or a reliable electrical disconnection between the PCB **570** and the ground portion (e.g., the ground terminal **4252***b*) of the antenna of the electronic device **501**.

In one or more example embodiments, the void portion **575** may be formed not in all the plurality of metal layers **571**. For example, the void portion **575** may not be formed in the first metal layer **571***a* that is located farthest from the coupling region **522** (e.g., one surface of the protruding rib **524**) among the plurality of metal layers **571**, but be formed on at least some of other metal layers, for example, **571***b* and **571***c*. This may ensure the plurality of lines **574** to pass through in the overlap region A**1** of the PCB **570** overlapping the coupling region **522** (e.g., one surface of the protruding rib **524**) where the burrs **5243** may occur and may thereby improve the wiring efficiency of the lines **574**.

In one or more example embodiments, the void portion **575** may be formed only on the third metal layer **571***c* that is located closest to the coupling region **522** (e.g., one surface of the protruding rib **524**). This may ensure a stable electrical connection or a reliable electrical disconnection between the PCB **570** and the ground portion (e.g., the ground terminal **4252***b*) of the antenna of the electronic device **501**, while improving the wiring efficiency of the plurality of lines **574** that passes above and/or below the plurality of metal layers **571**.

Referring to FIGS. **6**A to **6**C, according to one or more example embodiments, an electronic device **601** (e.g., the electronic device **401**) may include at least one housing **620** (e.g., the housing **420**), and the at least one housing **620** may include a plurality of side members **625** (e.g., the side members **425**) that forms a side frame defining an internal space of the housing **620**.

The electronic device **601** may include an antenna **680** that serves as a radiating conductor. A processor (e.g., the processor **120**) and/or a communication module (e.g., the communication module **190**) of the electronic device **601** may perform wireless communication through the antenna **680**. In one or more example embodiments, the antenna **680** may have a plate shape. In one or more example embodiments, the antenna **680** may be formed of a metal material. In one or more example embodiments, the antenna **680** may be electrically connected to one or more feed terminals **6251***a* and **6252***a* (e.g., the first feed terminal **4251***a* and/or the second feed terminal **4252***a*) formed on the plurality of side members **625**, and at least one ground terminal **6252***b* (e.g., the ground terminal **4252***b*). In one or more example embodiments, at least a portion of the antenna **680** may be coupled to a slit **626** (e.g., the slit **426**) between a pair of any neighboring side members **625**.

In one or more example embodiments, the housing **620** may include an accommodating portion **627** that accommodates therein the antenna **680**. The shape and size of the accommodating portion **627** may correspond to those of the antenna **680**.

In one or more example embodiments, a coupling region **622** (e.g., the coupling region **422**) of the housing **620** that overlaps an overlap region A**1** of a PCB **670** (e.g., the PCB **470**) may be located on a boundary **6271** of the accommodating portion **627** in which the antenna **680** is accommodated. This may be construed that a portion serving as the ground of the antenna **680** (e.g., a portion connected to the ground terminal **6252***b*) is required to be close to the antenna **680** and is required to be far away from a portion (e.g., the coupling region **622**) where a burr (e.g., the burr **5243**) may occur, which may ensure a stable electrical connection and/or a reliable electrical disconnection between the housing **620**, the PCB **670**, and the antenna **680**.

In addition, in one or more example embodiments, in a case in which the electronic device **601** is a foldable electronic device (e.g., the electronic device **201**), considering that a surface current density in areas of the antenna **680** located on the boundary **6271** of the accommodating portion **627** is large when the electronic device **601** is in a substantially unfolded state (e.g., as shown in FIG. **2**A) and the electronic device **601** is in a substantially folded state (e.g., as shown in FIG. **2**B), applying the structure of the PCB **570** described above with reference to FIG. **5**A to the coupling region **622** located on the boundary **6271** of the accommodating portion **627** may ensure a stable electrical connection and/or a reliable electrical disconnection and reduce a performance deviation of the antenna **680** related thereto.

Referring to FIG. **7**, according to one or more example embodiments, an electronic device **701** (e.g., the electronic device **501**) may include a housing **720** (e.g., the housing **520**) and a PCB **770** (e.g., the PCB **570**) coupled to the housing **720**. The housing **720** may include a protruding rib **724** (e.g., the protruding rib **524**), a hole **7241** (e.g., the hole **5241**), and a fixing member **7242** (e.g., the fixing member **5242**). A coupling region **722** (e.g., the coupling region **522**) may include a burr **7243** (e.g., the burr **5243**) formed on one surface (e.g., an upper surface) of the protruding rib **724**. The PCB **770** may include a plurality of metal layers **771** (e.g., the plurality of metal layers **571**), a plurality of printed solder resists **772** (e.g., the plurality of printed solder resists **572**), and a plurality of dielectrics **773** (e.g., the plurality of dielectric **573**). The PCB **770** may include an overlap region A**1** that overlaps the coupling region **722** of the housing **720** and a non-overlap region A**2** that does not overlap the coupling region **722**.

The PCB **770** may include a void portion **775** (e.g., the void portion **575**) in which a portion of at least one metal layer (e.g., at least one second metal layer **721***b* and/or a third metal layer **721***c*) close to the coupling region **722** among the plurality of metal layers **771** is not formed. In one or more example embodiments, the void portion **775** may be formed not in all the plurality of metal layers **771**. For example, the void portion **775** may not be formed in a first metal layer **771***a* but may be formed in at least one of other metal layers, for example, **771***b* and **771***c*. In one or more example embodiments, the void portion **775** may be formed only on the third metal layer **771***c* closest to the coupling region **722** (e.g., one surface of the protruding rib **724**).

In one or more example embodiments, the void portion **775** may be located throughout at least a portion of the overlap region A**1** and at least a portion of the non-overlap region A**2** of the PCB **770**. In one or more example embodiments, the void portion **775** may be located over substantially all portions of the overlap region A**1** and at least a portion of the non-overlap region A**2** of the PCB **770**. This structure, as in the structure of the PCB **570** described above with reference to FIGS. **5**A and **5**B, may prevent a direct contact between the burr **7243** and a line formed on at least one metal layer and may thereby ensure a stable electrical connection or a reliable electrical disconnection between the PCB **770** and a ground portion (e.g., the ground terminal **4252***b*) of an antenna of the electronic device **701**.

According to one or more example embodiments, an electronic device **501** may include a housing **520** having a coupling region **522**, and a PCB **570** having an overlap region A**1** that overlaps the coupling region **522** and a non-overlap region A**2** that does not overlap the coupling region **522**. The PCB **570** may include a plurality of metal layers **571** and a plurality of lines **574**, and may also include a void portion **575** in which a portion of at least one metal layer (e.g., **571***b* and **571***c*) close to the coupling region **522** among the plurality of metal layers **571** is not formed.

In one or more embodiments, the void portion **575** may not be located in the metal layer **571***a* that is farthest from the coupling region **522** among the plurality of metal layers **571**.

In one or more embodiments, the void portion **575** may be located only in the metal layer **571***c* that is closest to the coupling region **522** among the plurality of metal layers **571**, in the non-overlap region A**2**.

In one or more embodiments, the plurality of lines **574** may be formed in both the overlap region A**1** and the non-overlap region A**2**.

In one or more embodiments, an antenna **680** including a ground portion may be further included, and the ground portion may be electrically connected to the metal layer **571***c* closest to the coupling region **522** among the plurality of metal layers **571**.

In one or more example embodiments, an electronic device **601** may further include an antenna **680**, a housing **620** may include an accommodating portion **627** configured to accommodate the antenna **680**, and a coupling region **622** may be located on a boundary **6271** of the accommodating portion **627**.

In one or more example embodiments, the plurality of lines **574** may be lines through which signals pass.

In one or more example embodiments, the electronic device **601** may further include the antenna **680** including a ground portion, and the ground portion may be connected to the plurality of lines **574** or may form at least a portion of the plurality of lines **574**.

In one or more example embodiments, a housing **420** may have a bottom surface **421** having a coupling region **422** and a non-coupling region **423** different from the coupling region **422**, and the housing **420** may include a protruding rib (e.g., **424** or **524**) that protrudes from the coupling region (e.g., **422** or **522**) of the bottom surface **421** and contacts the metal layer **571***c* closest to the coupling region (e.g., **422** or **522**) among the plurality of metal layers **571**.

In one or more example embodiments, the housing **420** may include a hole **4241** formed in the protruding rib **424** and a fixing member **4242** coupled to the hole **4241** and configured to fix the PCB **470** to the protruding rib **424**.

In one or more example embodiments, the housing **520** may include a burr **5243** that is formed on the protruding rib **524** and meets the metal layer **571***c* closest to the coupling region **522** among the plurality of metal layers **571**.

In one or more example embodiments, the PCB **570** may further include at least one dielectric layer **573** located on or under at least one metal layer **571** among the plurality of metal layers **571**.

According to one or more example embodiments, an electronic device (e.g., **201** or **501**) may include a first housing **210** having a coupling region **522**, a second housing **220**, and a hinge structure connecting the first housing **210** and the second housing **220**, a display **261** including a first area **261***a* disposed on the first housing **210**, a second area **261***b* disposed on the second housing **220**, and a flexible area **261***c* between the first area **261***a* and the second area **261***b*, and a PCB **570** disposed on the first housing (e.g., **210** or **520**) and having an overlap region A**1** that overlaps the coupling region **522** and a non-overlap region A**2** that does not overlap the coupling region **522**. The PCB **570** may include a plurality of metal layers **571** and a plurality of lines **574**, and the PCB **570** may include a void portion **575** in which a portion of at least one metal layer (e.g., **571***b* and **571***c*) close to the coupling region **522** among the plurality of metal layers **571** is not formed, in the non-overlap region A**2**.

In one or more example embodiments, the void portion **575** may not be disposed in a metal layer **571***a* that is farthest from the coupling region **522** among the plurality of metal layers **571**.

In one or more example embodiments, the void portion **575** may be disposed only in the metal layer **571***c* that is closest to the coupling region **522** among the plurality of metal layers **571**, in the non-overlap region A**2**.

In one or more example embodiments, the plurality of lines **574** may be formed in both the overlap region A**1** and the non-overlap region A**2**.

In one or more example embodiments, an electronic device (e.g., **501** or **601**) may further include an antenna **680** including a ground portion, and the ground portion may be electrically connected to the metal layer **571***c* closest to the coupling region **522** among the plurality of metal layers **571**.

In one or more example embodiments, the electronic device (e.g., **501** or **601**) may further include the antenna **680**, a housing **620** may include an accommodating portion **627** configured to accommodate the antenna **680**, and a coupling region **622** may be located on a boundary **6271** of the accommodating portion **627**.

According to one or more example embodiments, an electronic device **701** may include a housing **720** having a coupling region **722**, and a PCB **770** having an overlap region A**1** that overlaps the coupling region **722** and a non-overlap region A**2** that does not overlap the coupling region **722**. The PCB (e.g., **570** or **770**) may include a plurality of metal layers (e.g., **571** or **771**) and a plurality of lines **574**, and the PCB **770** may include a void portion **775** in which a portion of at least one metal layer (e.g., **771***b* and/or **771***c*) close to the coupling region **722** among the plurality of metal layers **771** is not formed.

In one or more embodiments, the void portion **775** may be disposed over at least a portion of the overlap region A**1** and at least a portion of the non-overlap region A**2**.

While various example embodiments have been particularly shown and described, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

What is claimed is:

1. An electronic device comprising:
a housing comprising a coupling region; and
a printed circuit board (PCB) comprising:
an overlap region that overlaps the coupling region;
a non-overlap region that does not overlap the coupling region;
a plurality of metal layers comprising a plurality of lines; and
a void portion in which a portion of at least one metal layer, among the plurality of metal layers, is not formed so that the at least one metal layer is discontinuous or terminated in the void portion, the at least one metal layer including a metal layer that is closest to the coupling region among the plurality of metal layers.

2. The electronic device of claim **1**, wherein the at least one metal layer does not include a metal layer that is farthest from the coupling region among the plurality of metal layers.

3. The electronic device of claim **1**, wherein the void portion is in the non-overlap region, and the at least one metal layer includes only the metal layer that is closest to the coupling region.

4. The electronic device of claim **1**, wherein the plurality of lines is provided in both the overlap region and the non-overlap region.

5. The electronic device of claim **1**, further comprising:
an antenna comprising a ground portion,
wherein the ground portion is electrically connected to the metal layer that is closest to the coupling region among the plurality of metal layers.

6. The electronic device of claim **1**, further comprising:
an antenna,
wherein the housing comprises an accommodating portion configured to accommodate the antenna,
wherein the coupling region is at a boundary of the accommodating portion.

7. The electronic device of claim **1**, wherein the plurality of lines are lines through which signals pass.

8. The electronic device of claim **1**, further comprising:
an antenna comprising a ground portion that is connected to the plurality of lines or forms at least a portion of the plurality of lines.

9. The electronic device of claim **1**, wherein the housing has a bottom surface comprising the coupling region and a non-coupling region that is different from the coupling region, and
wherein the housing further comprises a protruding rib that protrudes from the coupling region of the bottom surface and contacts the PCB.

10. The electronic device of claim **9**, wherein the housing further comprises:
a hole formed on the protruding rib; and
a fixing member coupled to the hole and configured to fix the PCB to the protruding rib.

11. The electronic device of claim **9**, wherein the protruding rib comprises a burr contacting the metal layer that is closest to the coupling region.

12. The electronic device of claim **1**, wherein the PCB further comprises at least one dielectric layer on or under at least one metal layer among the plurality of metal layers.

13. The electronic device of claim **1**, wherein the void portion is in the non-overlap region.

14. The electronic device of claim **1**, wherein the void portion is in at least a portion of the overlap region and at least a portion of the non-overlap region.

15. The electronic device of claim **1**, wherein the housing further comprises a first housing comprising the coupling region and a second housing, and
wherein the electronic device comprises:
a hinge structure connecting the first housing and the second housing; and
a display comprising a first area disposed in the first housing, a second area disposed in the second housing, and a flexible area between the first area and the second area.

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